

Notebook for EDA

1. Read in cleaned data files

```
In [136... import pandas as pd
import numpy as np
import altair as alt
import seaborn as sns
import matplotlib.pyplot as plt
```

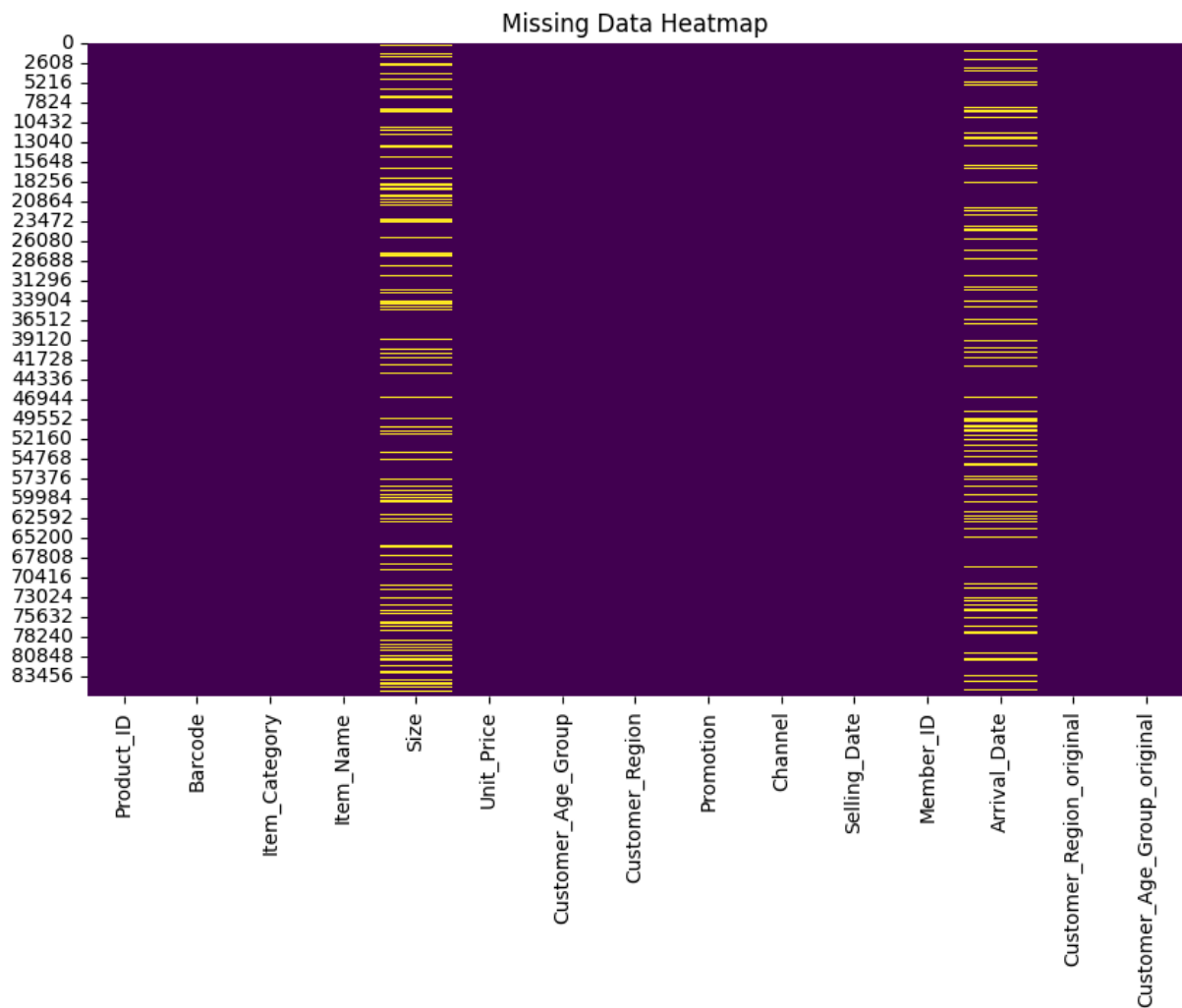
```
In [137... stadium = pd.read_csv('data/processed/clean_stadium.csv')
fanbase = pd.read_csv('data/processed/fanbase_clean.csv')
merch = pd.read_csv('data/processed/cleaned_merch.csv')
fan_merch_merged = pd.read_csv('data/processed/merch_fanbase_merged.csv')
```

/var/folders/wp/pxd2hj7s5j1ckh6z7fz4mwkm0000gn/T/ipykernel_69083/4120701562.py:4: DtypeWarning:

Columns (2,3,4,6,7,8,10) have mixed types. Specify dtype option on import or set low_memory=False.

Missing values in Merch

```
In [138... plt.figure(figsize=(10, 6))
sns.heatmap(merch.isnull(), cbar=False, cmap='viridis')
plt.title("Missing Data Heatmap")
plt.show()
```



Missing At Random (MAR):

I already know from looking at the data that missing values in arrival date align with merch bought in store, and missing sizes align with products that do not require sizes (such as caps, mugs, posters) but this would be good to confirm.

```
In [139... data = {
    'Product': merch['Product_ID'].unique(),
    'Item': merch['Item_Name'].unique()
}
product_table = pd.DataFrame(data)
product_table
```

Out [139...

	Product	Item
0	10000000	Classic Cap
1	10000001	Retro Cap
2	10000002	Team Hoodie
3	10000003	Training Hoodie
4	10000004	Away Jersey
5	10000005	Home Jersey
6	10000006	Retro Jersey
7	10000007	Club Mug
8	10000008	Logo Mug
9	10000009	Player Poster
10	10000010	Team Poster
11	10000011	Away Scarf
12	10000012	Home Scarf
13	10000013	Youth Team Hoodie
14	10000014	Youth Training Hoodie
15	10000015	Youth Away Jersey
16	10000016	Youth Home Jersey
17	10000017	Youth Retro Jersey

```
In [140... size_ids = [10000002, 10000003, 10000004, 10000005, 10000006, 10000013, 10000014]
merch['should_have_size'] = merch['Product_ID'].isin(size_ids)
merch['size_missing'] = merch['Size'].isna()
```

```
In [141... mismatches = merch[merch['should_have_size'] & merch['size_missing']]
mismatches.shape
```

Out [141... (0, 17)

OK, we confirm that all missing values in the size column are due to products not needing sizes.

```
In [142... merch = merch.drop(['size_missing', 'should_have_size'], axis=1)
```

Confirm that all missing values in arrival date are due to products being bought in store and not online:

```
In [143... merch['arrival_missing'] = merch['Arrival_Date'].isna()
merch['should_have_arrival'] = merch['Channel'] != 'Team Store'
```

```
In [144... arrival_mismatches = merch[merch['arrival_missing'] & merch['should_have_arrival_mismatches.shape
```

```
Out[144... (0, 17)
```

```
In [145... store_date_mismatches = merch[~merch['arrival_missing'] & (merch['Channel'] store_date_mismatches.shape
```

```
Out[145... (0, 17)
```

```
In [146... merch = merch.drop(['arrival_missing', 'should_have_arrival'], axis=1) merch.head()
```

```
Out[146... Product_ID Barcode Item_Category Item_Name Size Unit_Price Customer_Ag
```

0	10000000	20000000	Cap	Classic Cap	NaN	35
---	----------	----------	-----	-------------	-----	----

1	10000000	20000001	Cap	Classic Cap	NaN	35
---	----------	----------	-----	-------------	-----	----

2	10000000	20000002	Cap	Classic Cap	NaN	35
---	----------	----------	-----	-------------	-----	----

3	10000000	20000003	Cap	Classic Cap	NaN	35
---	----------	----------	-----	-------------	-----	----

4	10000000	20000004	Cap	Classic Cap	NaN	35
---	----------	----------	-----	-------------	-----	----

Ok, we have confirmed that all missing values are logical and values that should have input, do have input

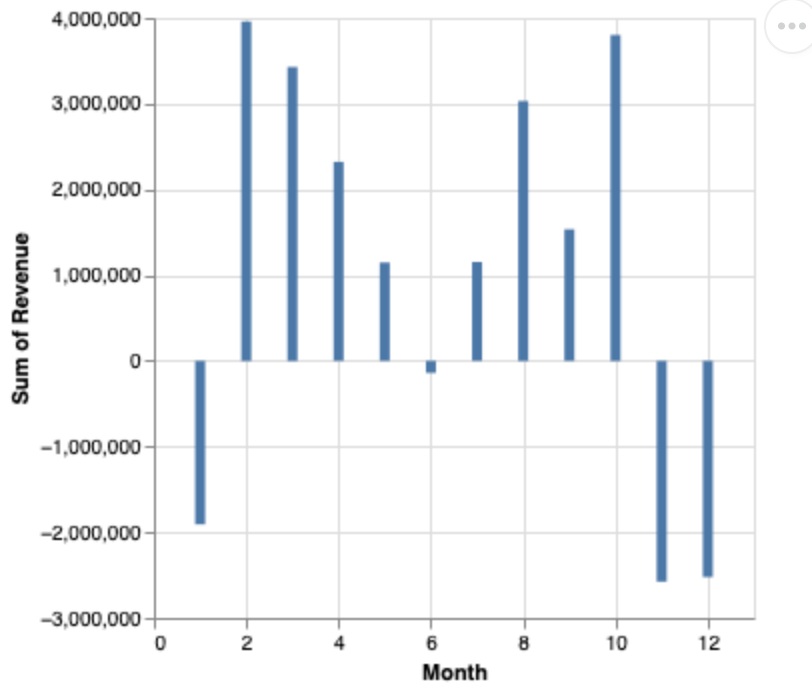
EDA: Stadium

1. Revenue Decomposition by Month:

- potentially reflects seasonality of games
- revenue loss months likely due to maintenace costs not being balanced out by ticket/merch/food sales
- interesting V shaped trend with June being a revenue loss month
 - We should look into this as an avenue to mitigate loss

```
In [147... overall_revenue = alt.Chart(stadium).mark_bar().encode(  
    x='Month',  
    y='sum(Revenue) ',  
)  
overall_revenue
```

Out [147...

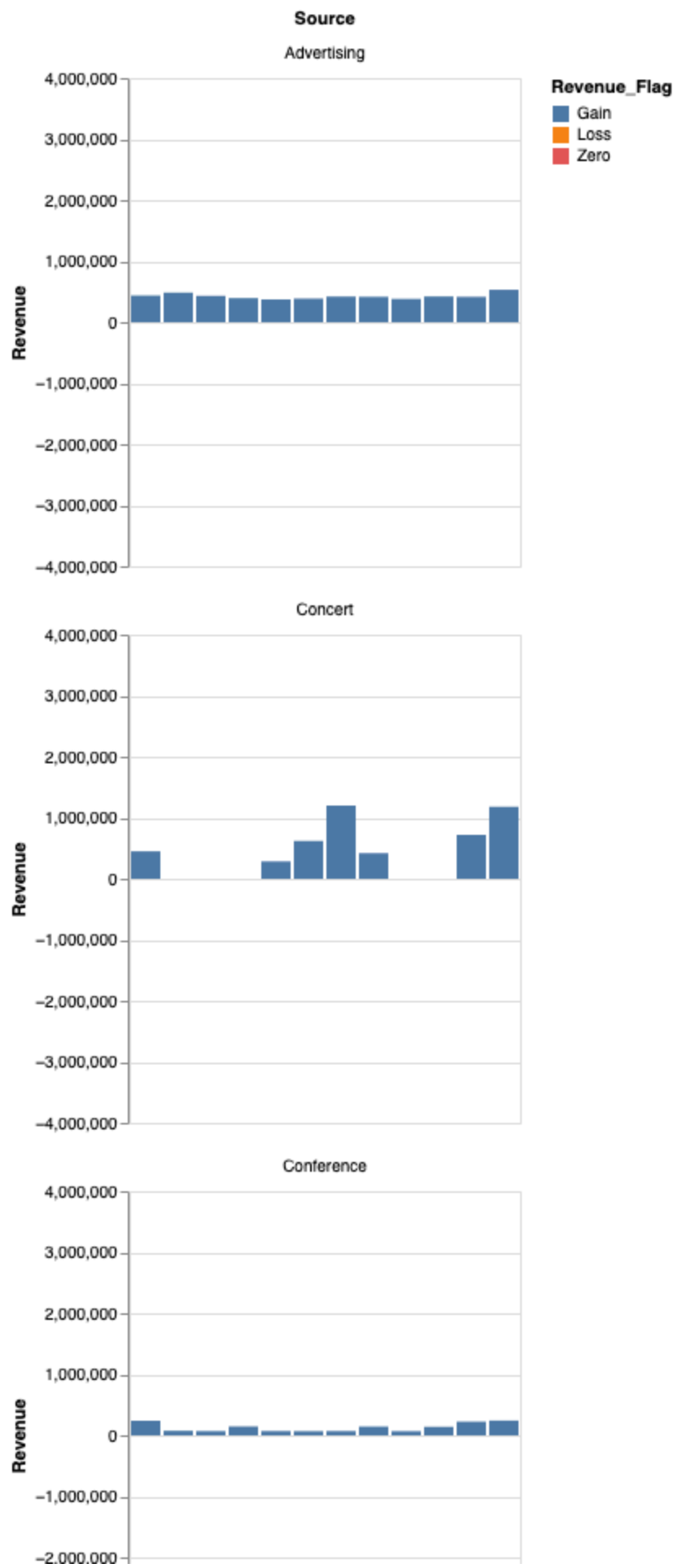


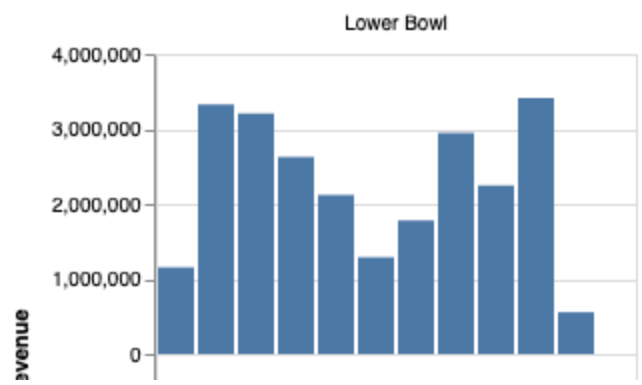
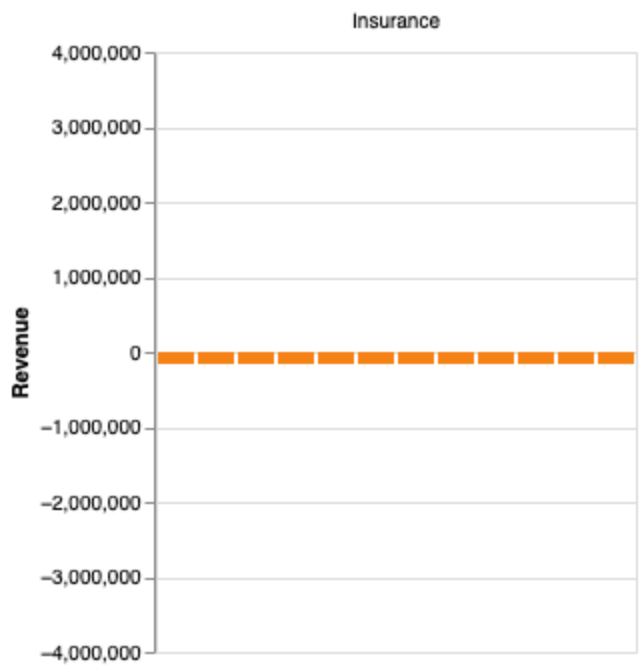
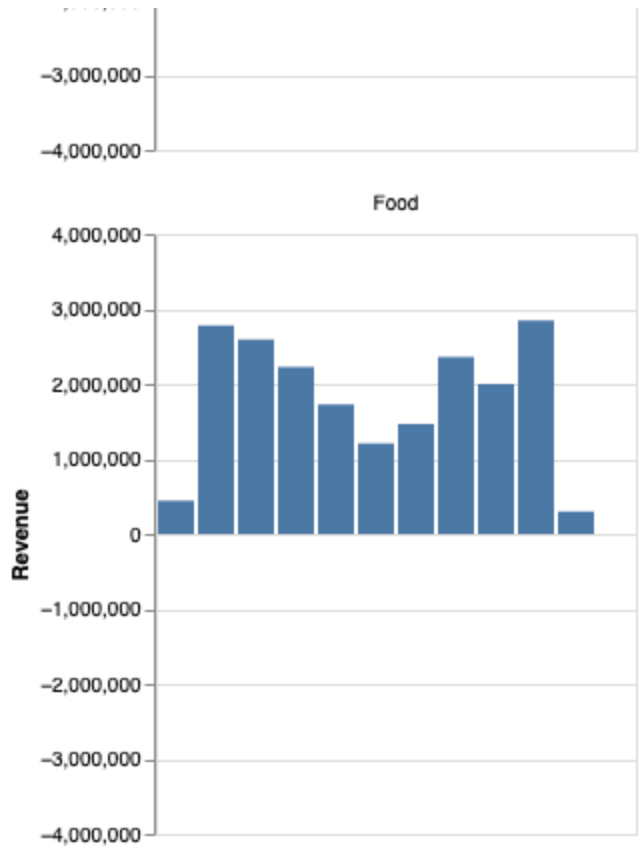
2. Revenue breakdown by Source (monthly)

- sources of loss: Maintenance, Insurance, Staff, Utilities
- main sources of gain: Upper Bowl ticket sales, food
- potential sources to explore: Concerts (seems to line up with off season)

In [148...

```
rev_gain_or_loss = alt.Chart(stadium).mark_bar().encode(  
    x='Month:O',  
    y='Revenue:Q',  
    color='Revenue_Flag:N',  
).facet(  
    'Source',  
    columns = 1  
)  
rev_gain_or_loss
```





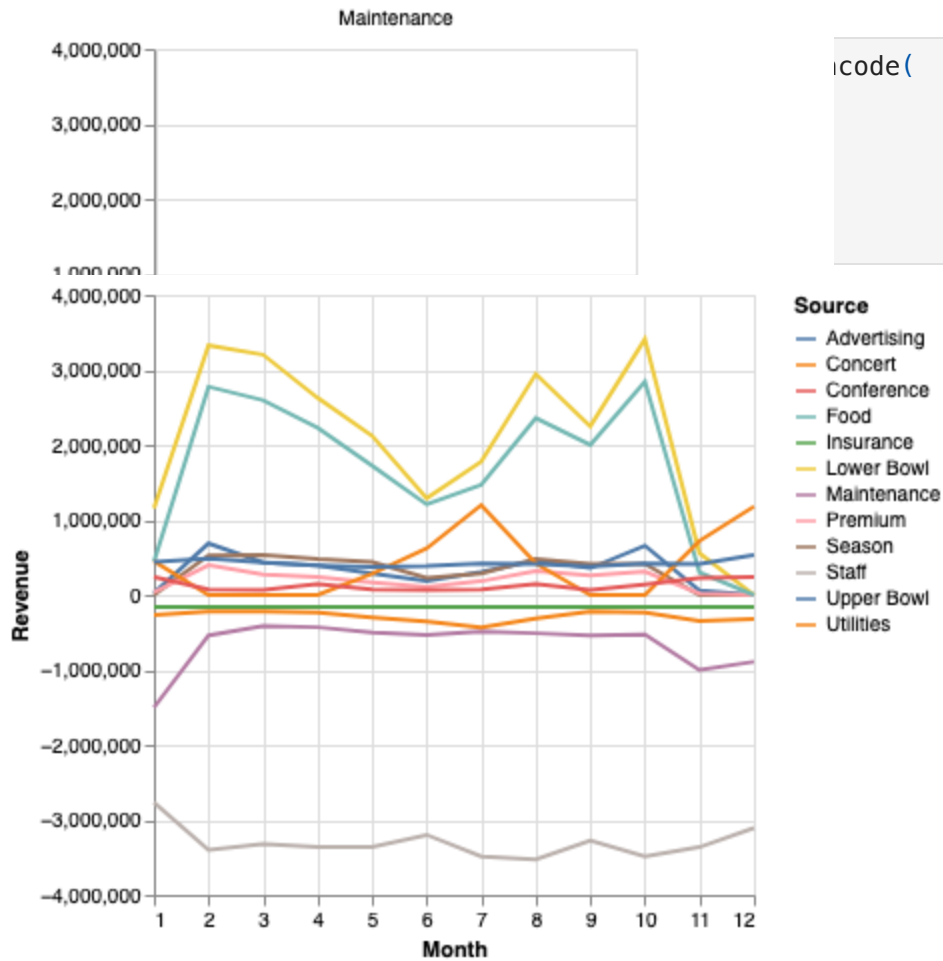


ember

In [149...

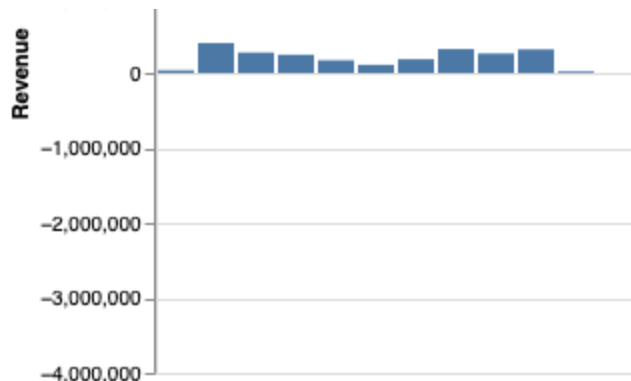
code(

Out [149...



Source

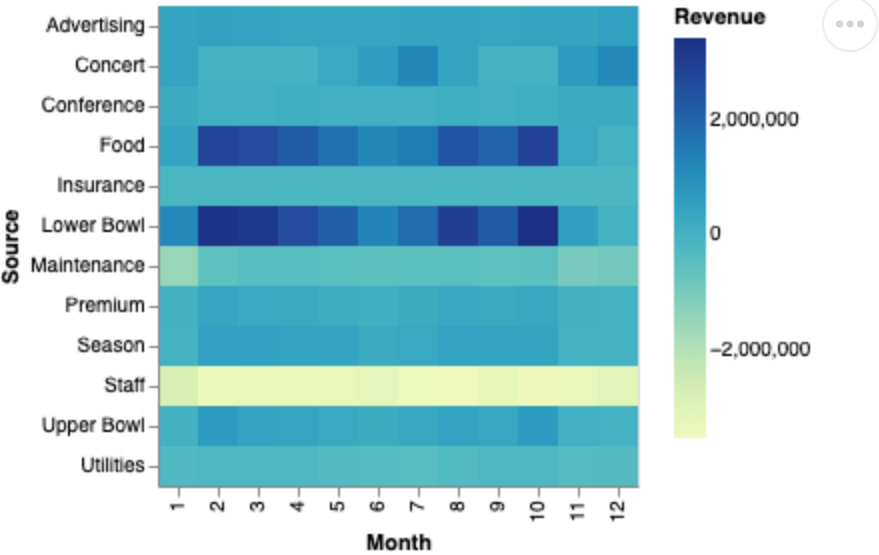
- Advertising
- Concert
- Conference
- Food
- Insurance
- Lower Bowl
- Maintenance
- Premium
- Season
- Staff
- Upper Bowl
- Utilities



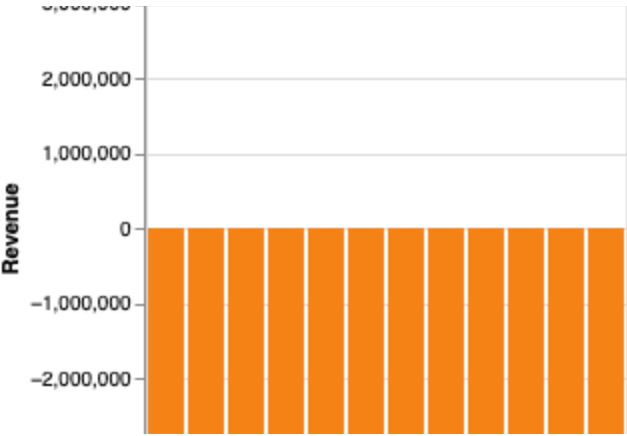
In [150...



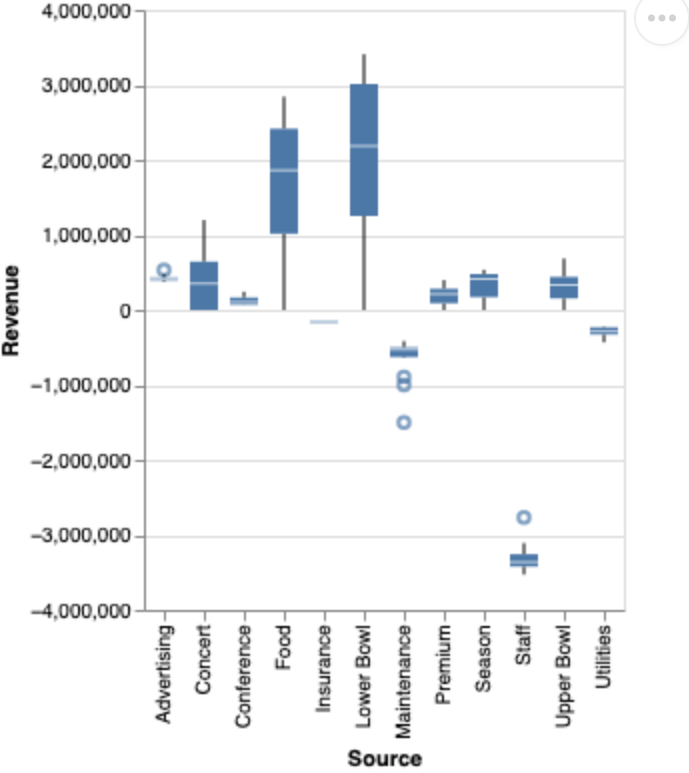
Out [150...



In [151...



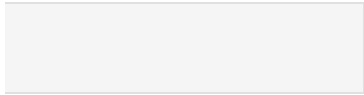
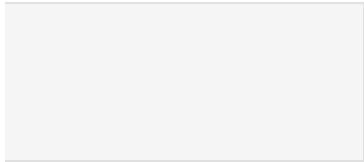
Out [151...



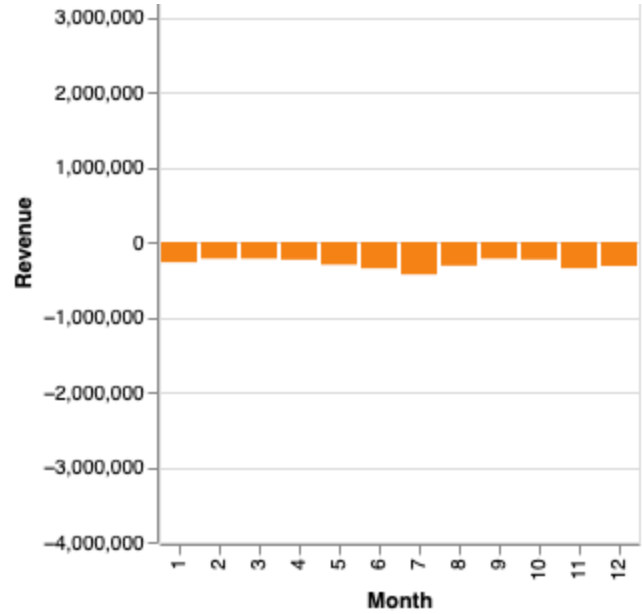
In [152...



le to seasonal peaks
read of data



Out [152...
In [153...
Out [153...
In [154...
Out [154...



```
agg({'Revenue': 'sum'})  
, inplace=True)
```

3	Food	19948319
4	Insurance	-1920000
5	Lower Bowl	24669304
6	Maintenance	-7870000
7	Premium	2322177
8	Season	3807502
9	Staff	-39646000
10	Upper Bowl	3855705
11	Utilities	-3469941

Insights:

- primary sources of income are from Lower Bowl and Food
 - Upper Bowl is significantly lower earning than LB
 - The highest source of loss is Staff
 - February and October have the overall highest profit
 - January, June, November, December are overall losses
 - insurance seems to be the only fixed cost (arguably, staff as well, but it does vary month by month)
1. The overall revenue across the 2024 period was: \$13,233,516.00
 2. The highest earning month was February with: \$3'956'727.00
 3. The lowest earning month was November with: -\$2'576'211.00 (followed closely by December)
 4. Total staff costs were: -\$39'646'000.00
 5. Total Lower Bowl earnings were: \$24'669'304.00

EDA: Fanbase

```
In [155... alt.data_transformers.disable_max_rows()
```

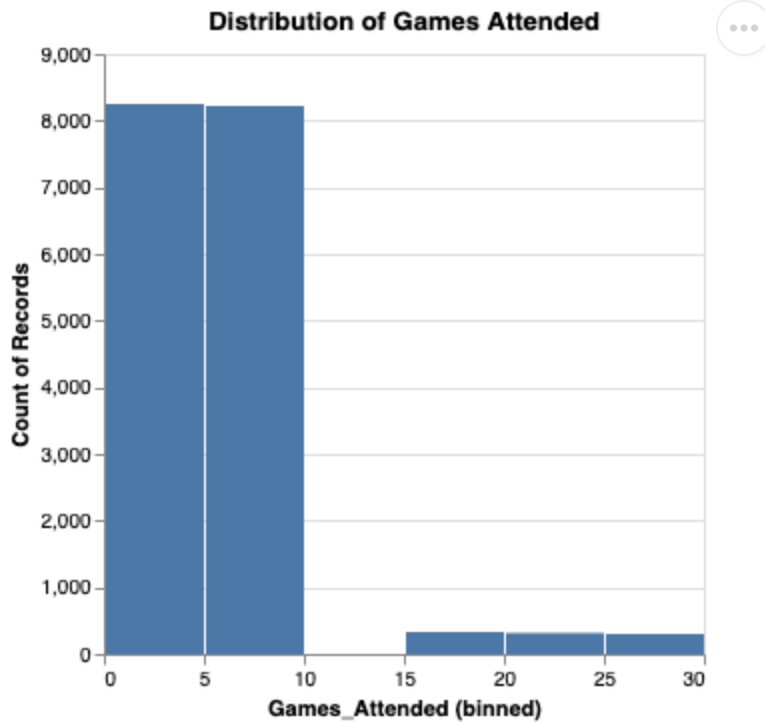
```
Out[155... DataTransformerRegistry.enable('default')
```

1. Games Attended total distributions

- most fans attend between 0-10 games, very few attend 15-30
- no one attended between 10-15 games?

```
In [156... alt.Chart(fanbase).mark_bar().encode(  
    x=alt.X('Games_Attended:Q', bin=True),  
    y='count()',  
    tooltip=['Games_Attended']  
) .properties(title='Distribution of Games Attended')
```

Out [156...

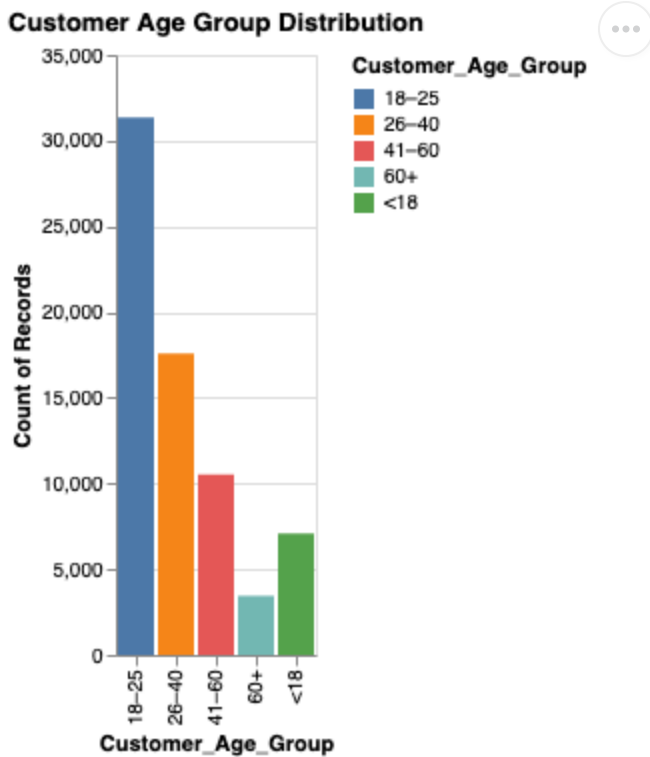


2. Distribution of fanbase membership by age group

In [157...

```
alt.Chart(fanbase).mark_bar().encode(  
    x='Customer_Age_Group:N',  
    y='count()',  
    color='Customer_Age_Group:N'  
).properties(title='Customer Age Group Distribution')
```

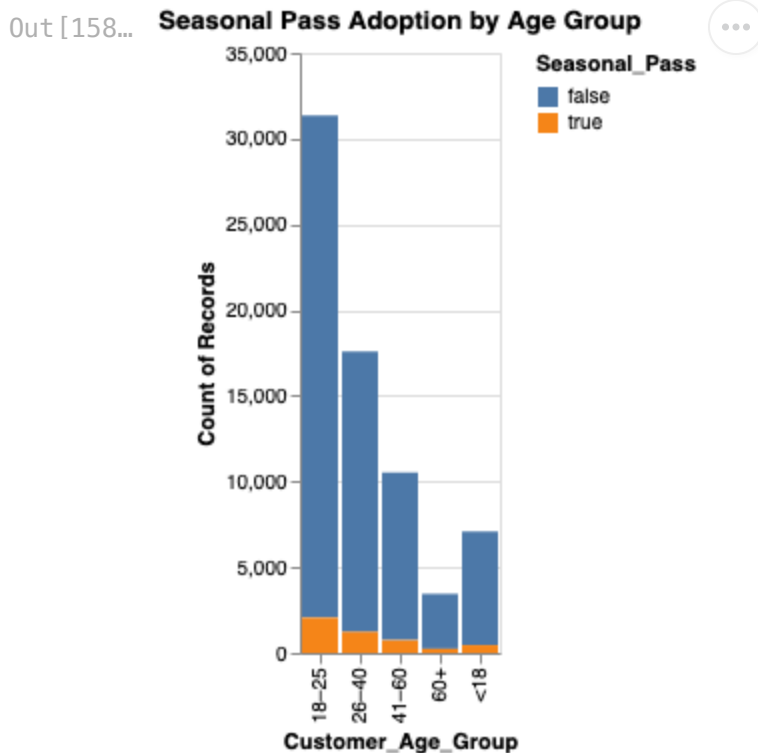
Out [157...



3. Seasonal pass adoption, by age group

- no apparent trends, proportional to density of each age group

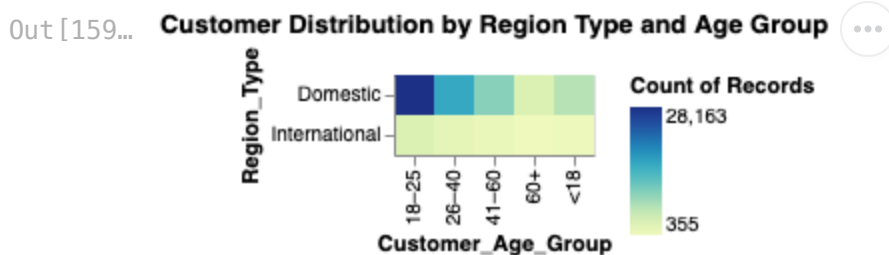
```
In [158... alt.Chart(fanbase).mark_bar().encode(  
    x='Customer_Age_Group:N',  
    y='count()',  
    color='Seasonal_Pass:N'  
) .properties(title='Seasonal Pass Adoption by Age Group')
```



4. Heatmap of age group vs region belonging

- the vast majority of fans are domestic, across all age groups

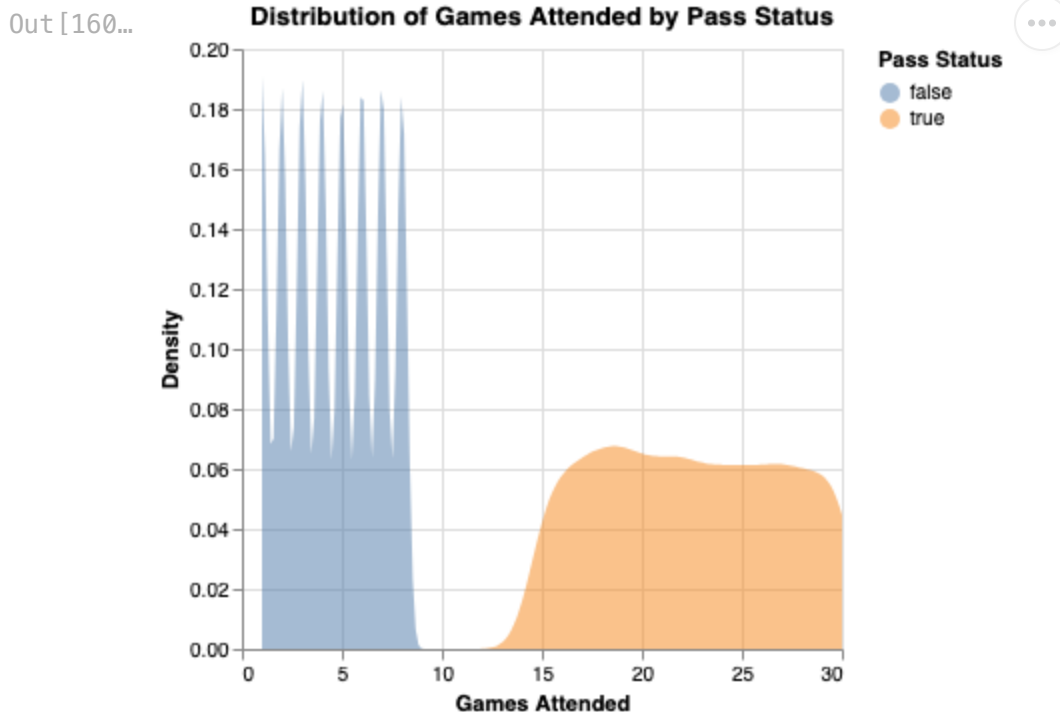
```
In [159... alt.Chart(fanbase).mark_rect().encode(  
    x='Customer_Age_Group:N',  
    y='Region_Type:N',  
    color='count()',  
    tooltip=['Customer_Age_Group', 'Region_Type']  
) .properties(title='Customer Distribution by Region Type and Age Group')
```



5. Density of pass holders and game attendance

- no overlap at all! If they are a season pass holder they see between 15-30 games
- explains the bimodality in the earlier histogram

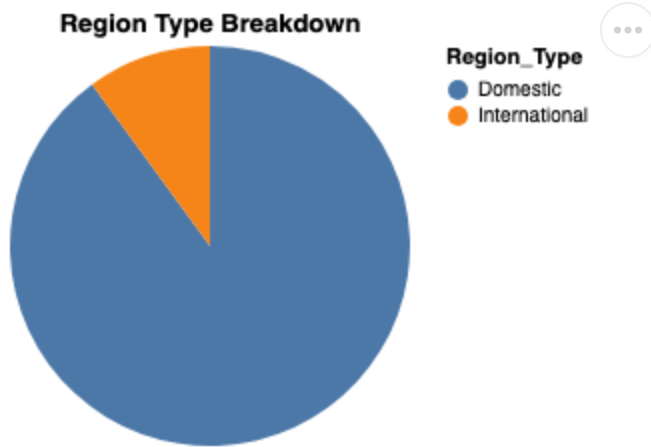
```
In [160... alt.Chart(fanbase).transform_density(  
    density='Games_Attended',  
    groupby=['Seasonal_Pass'],  
    as_=['Games_Attended', 'density']  
)  
.mark_area(opacity=0.5).encode(  
    x=alt.X('Games_Attended:Q', title='Games Attended'),  
    y=alt.Y('density:Q', title='Density'),  
    color=alt.Color('Seasonal_Pass:N', title='Pass Status')  
)  
.properties(  
    title='Distribution of Games Attended by Pass Status',  
)
```



6. Region Type composition

```
In [161... alt.Chart(fanbase).mark_arc().encode(  
    theta='count()',  
    color='Region_Type:N',  
    tooltip=['Region_Type']  
)  
.properties(  
    title='Region Type Breakdown',  
    width=200,  
    height=200  
)
```

Out [161...

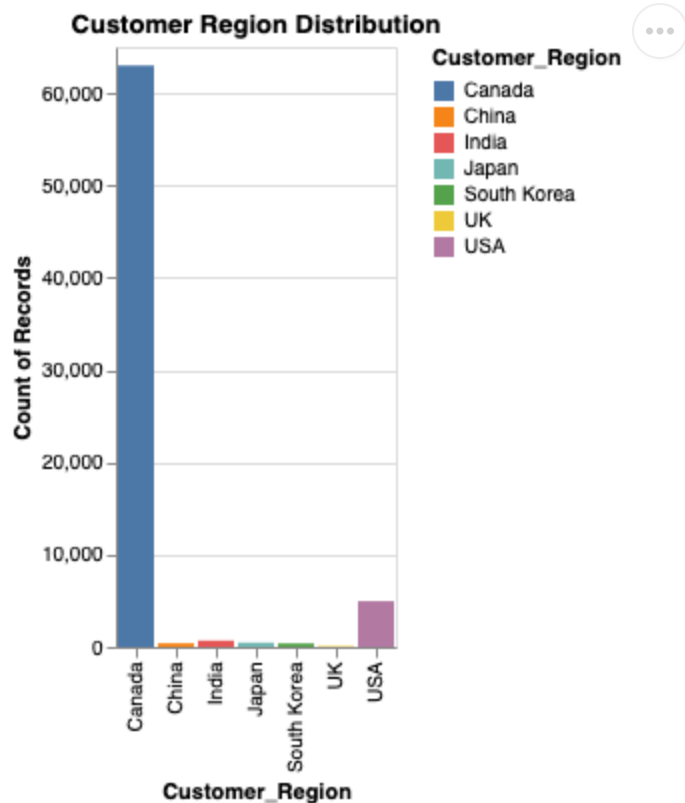


7. Specific customer region distribution

In [162...

```
alt.Chart(fanbase).mark_bar().encode(  
    x='Customer_Region:N',  
    y='count()',  
    color='Customer_Region:N'  
) .properties(title='Customer Region Distribution')
```

Out [162...



In [163...

```
fanbase['Seasonal_Pass'].value_counts()
```

Out [163...

```
Seasonal_Pass  
False    65251  
True      4749  
Name: count, dtype: int64
```

Insights:

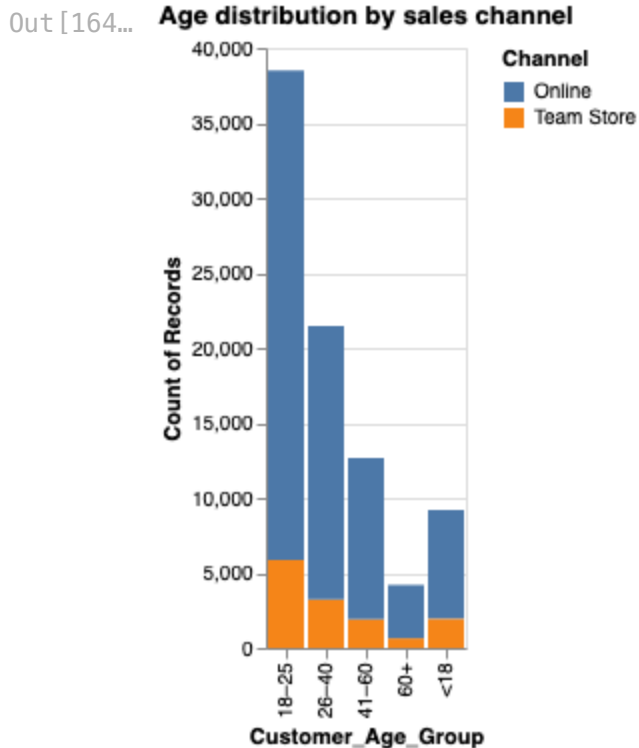
- The vast majority of memberships come from Domestic fans
- The largest demographic of fans are 18-25 year olds
- Most people attend 0-10 games and are non season pass holders
- People who see 15-20 games are more likely to be season pass holders
- Seasonal pass holders make up about 7% of the fanbase
- Of the international fans - the majority come from the US, followed by India

EDA: Merch

1. Age distribution of merch purchases, plus usage of channels by age group

- as expected, largest group is 18-25 year olds
- online vs team store usage is proportional to demographic quantity

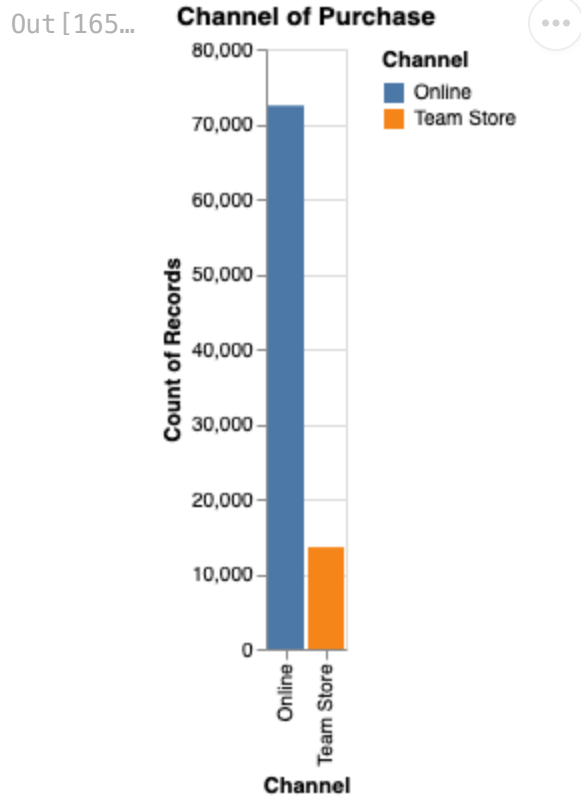
```
In [164... alt.Chart(merch).mark_bar().encode(  
    x='Customer_Age_Group',  
    y='count()',  
    color='Channel:N'  
) .properties(title='Age distribution by sales channel')
```



2. Channel Performance

- Vast majority purchase online

```
In [165... alt.Chart(merch).mark_bar().encode(  
    x='Channel:N',  
    y='count()',  
    color='Channel:N'  
) .properties(title='Channel of Purchase')
```



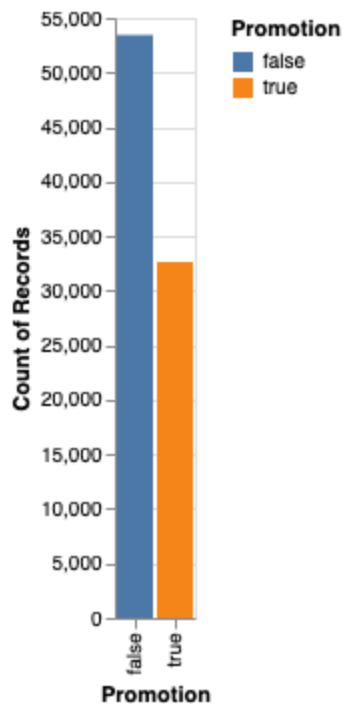
3. Purchases through promotion and without

- most are without, but a fair amount of purchases happen through a promotion

```
In [166... alt.Chart(merch).mark_bar().encode(  
    x='Promotion:N',  
    y='count()',  
    color='Promotion:N'  
) .properties(title='Distribution of Promotion Based Purchases')
```

Out [166...

Distribution of Promotion Based Purchases



4. Sales by item, including breakdown by channel and whether or not purchased with a promotion

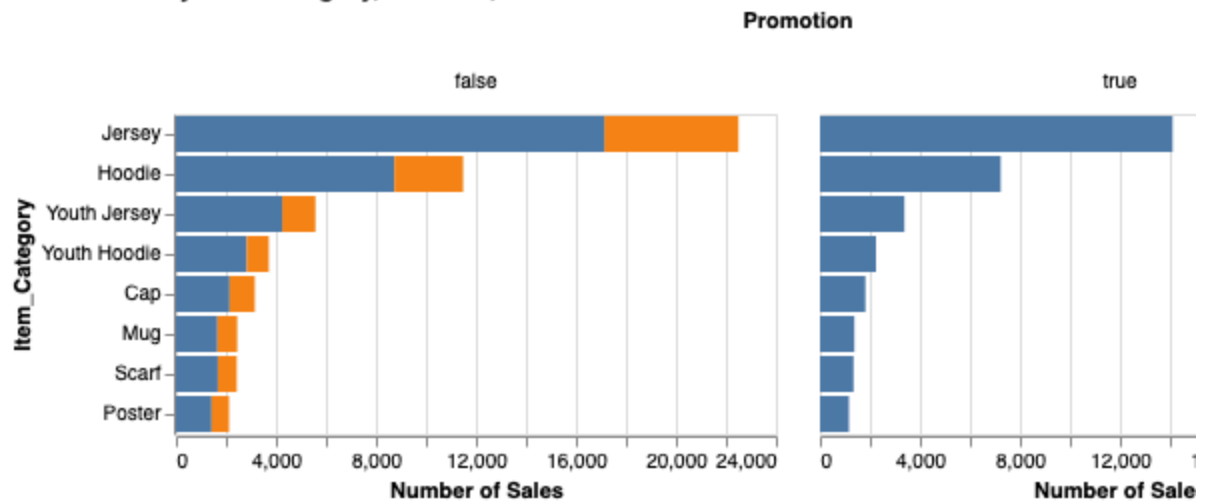
- Jerseys sell the most, followed by hoodies
- promotions are only used with online purchases

In [167...

```
alt.Chart(merch).mark_bar().encode(  
  x=alt.X('count()', title='Number of Sales'),  
  y=alt.Y('Item_Category:N', sort='-x'),  
  color='Channel',  
  column=alt.Column('Promotion:N', title='Promotion')  
)  
.properties(  
  title='Sales count by Item Category, Channel, and Promotion'  
)
```

Out [167...

Sales count by Item Category, Channel, and Promotion

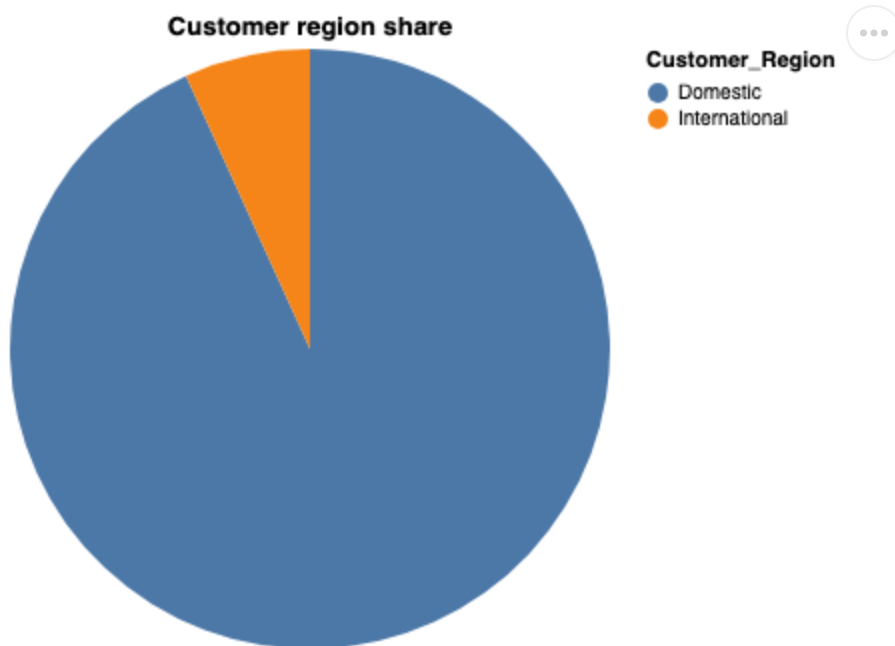


5. Customer Region

In [168...

```
alt.Chart(merch).mark_arc().encode(  
  theta='count()',  
  color='Customer_Region:N',  
) .properties(title='Customer region share')
```

Out [168...



6. Purchases through a promotion by age group

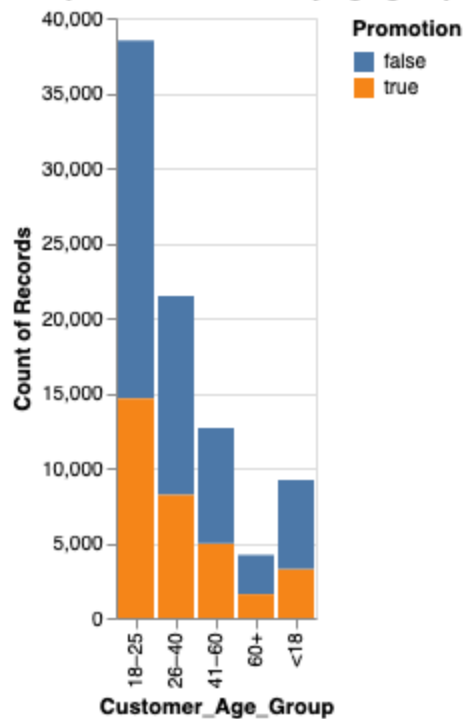
- proportional, no obvious trends

In [169...

```
alt.Chart(merch).mark_bar().encode(  
  x='Customer_Age_Group:N',  
  y='count()',  
  color='Promotion:N',  
) .properties(title='Promotional vs non-promotional sales by age group and ch')
```

Out [169...

Promotional vs non-promotional sales by age group and channel



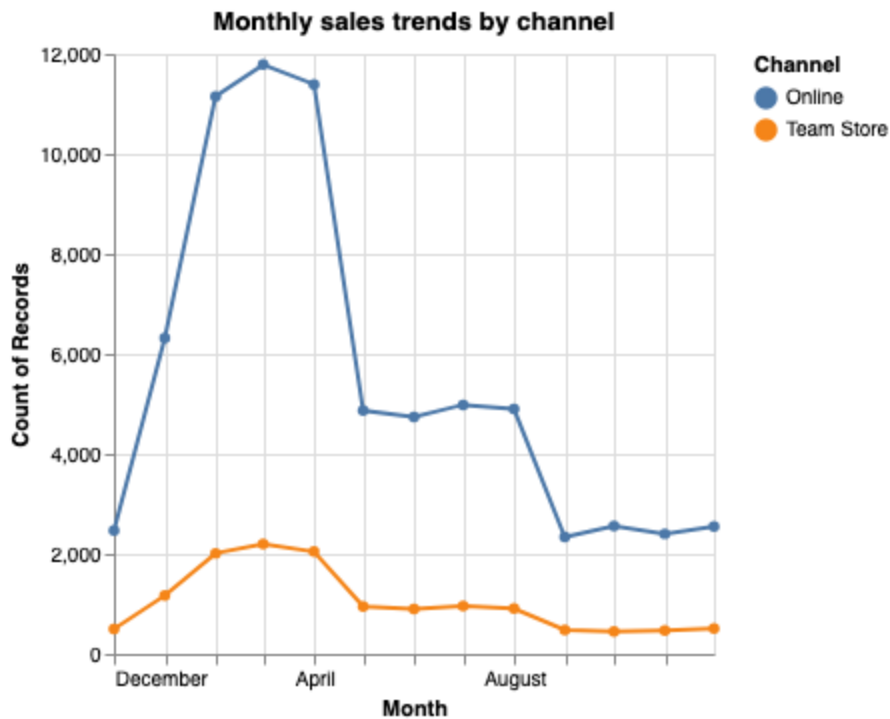
```
In [170... merch['Selling_Date'] = pd.to_datetime(merch['Selling_Date'])
merch['Month'] = merch['Selling_Date'].dt.to_period('M').astype(str)
```

7. Monthly sales trends

- peak sales occur in March - could be due to beginning of season
- dip in May and in September

```
In [171... alt.Chart(merch).mark_line(point=True).encode(
    x='Month:T',
    y='count()',
    color='Channel:N'
).properties(title='Monthly sales trends by channel')
```

Out [171...



In [172...

```
merch['Arrival_Date'] = pd.to_datetime(merch['Arrival_Date'])
merch['Selling_Date'] = pd.to_datetime(merch['Selling_Date'])
merch['delivery_days'] = (merch['Arrival_Date'] - merch['Selling_Date']).dt.
```

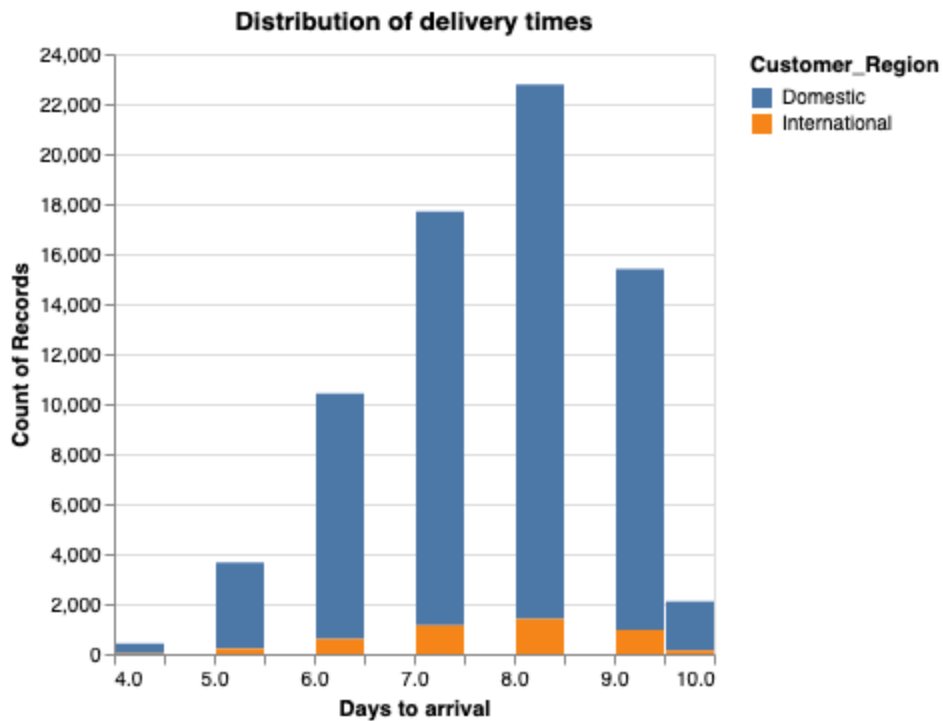
8. How long it usually takes to deliver items (in days)

- most items take about 8 days to be delivered
- some delays occurring with 9-10 days shipping time
- no obvious discrepancies between domestic and international arrival dates

In [173...

```
alt.Chart(merch.dropna(subset=['delivery_days'])).mark_bar().encode(
    x=alt.X('delivery_days:Q', bin=alt.Bin(maxbins=20), title='Days to arriv
    y='count()',
    color='Customer_Region'
).properties(title='Distribution of delivery times')
```

Out [173...

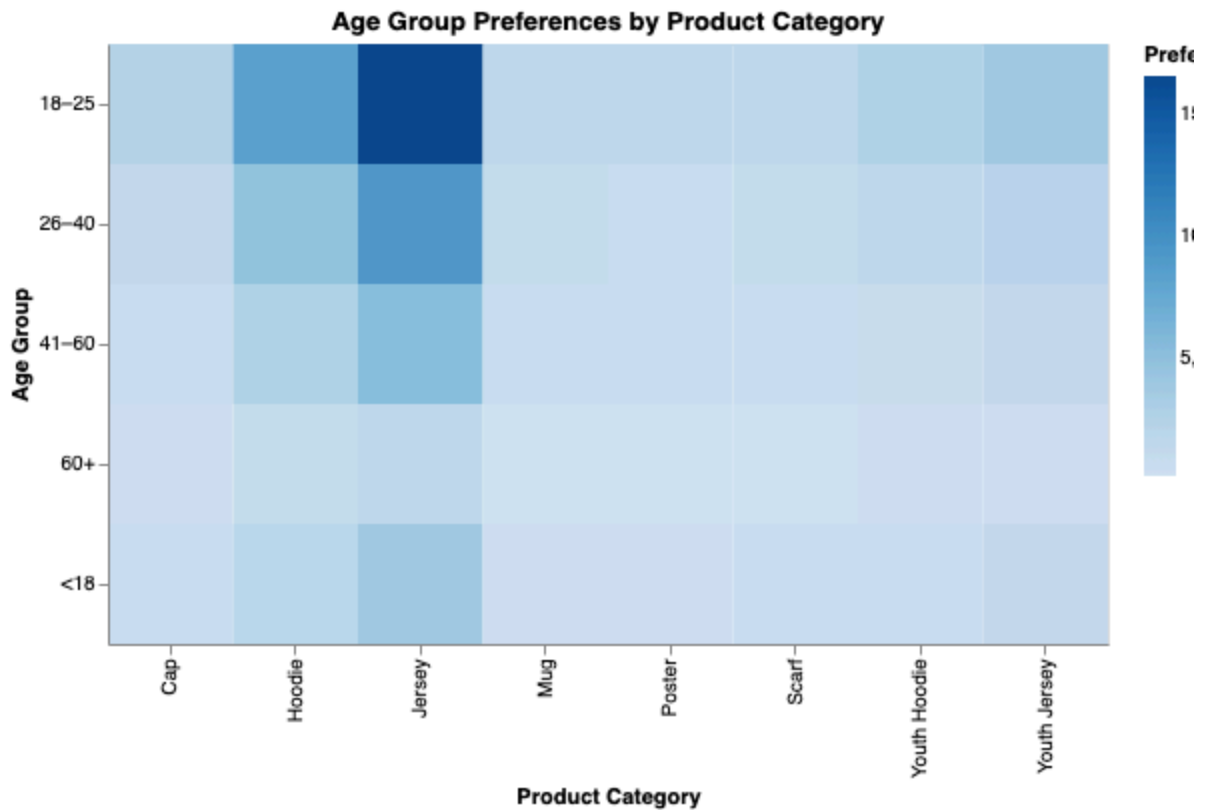


9. Heatmap of product preference by age group

- Jerseys are the most popular item for all age groups
- no age group specific trends

```
In [174... heatmap_data = merch.groupby(['Customer_Age_Group', 'Item_Category']).size()
heatmap_data_reset = heatmap_data.reset_index().melt(id_vars='Customer_Age_G
alt.Chart(heatmap_data_reset).mark_rect().encode(
    x=alt.X('Item_Category:N', title='Product Category'),
    y=alt.Y('Customer_Age_Group:N', title='Age Group'),
    color=alt.Color('Volume:Q', scale=alt.Scale(scheme='blues'), title='Pref
    tooltip=['Customer_Age_Group', 'Item_Category', 'Volume']
).properties(
    title='Age Group Preferences by Product Category',
    width=500,
    height=300
)
```

Out [174...



Insights:

- 18-25 year olds are the largest demographic for merch purchases, followed by 26-40 year olds*
- the majority of purchases happen online
- majority of purchases do not happen with a promotion, but a reasonably large amount are through a promotion
- all purchases through a promotion happen online (makes sense if promotions are online ads)
- the vast majority of merch purchases occur domestically*
- there does not seem to be a trend between age demographic and use of promotions
- major peak in purchases around March, with a dip in September
- most purchases take around 8 days to be delivered, with some delays leading to 9-10 day delivery

*note these values have been transformed based on the assumptions made in the data cleaning - exact values may not be accurate

EDA merged merch/fanbase data

1. Venn Diagram of fans who attended a game and purchased merch

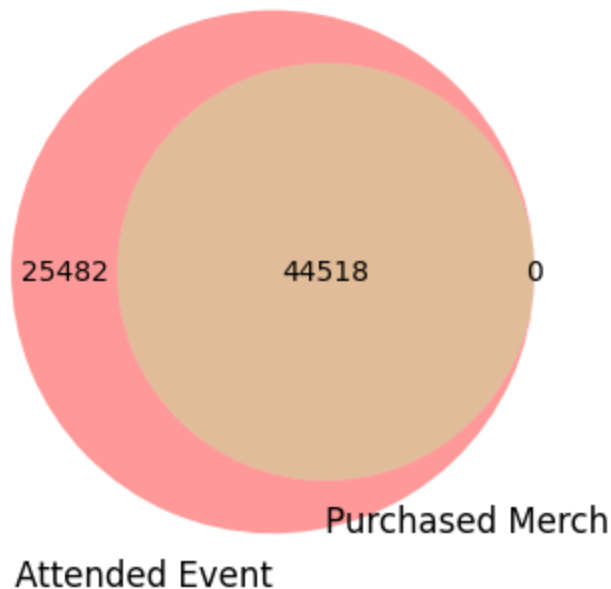
- promising for generating probability statements

```
In [176... from matplotlib_venn import venn2
import matplotlib.pyplot as plt

# Define sets based on Member_ID
attended = set(fan_merch_merged.loc[fan_merch_merged['Games_Attended'].notna])
purchased = set(fan_merch_merged.loc[fan_merch_merged['Item_Category'].notna])

# Plot Venn diagram
plt.figure(figsize=(6, 4))
venn2([attended, purchased], set_labels=('Attended Event', 'Purchased Merch'))
plt.title('Overlap Between Attendance and Merchandise Purchase')
plt.show()
```

Overlap Between Attendance and Merchandise Purchase



- ### 2. Calculating % of members who attending 0 games: 0! All member attended at least 1 game

```
In [ ]: total_members = fan_merch_merged['Member_ID'].nunique()

inactive_members = fan_merch_merged[(fan_merch_merged['Games_Attended'].isna())]

inactive_pct = (inactive_members / total_members) * 100
print(f"{inactive_pct:.2f}% of members attended 0 games.")
```

0.00% of members attended 0 games.

- ### 3. Trying to find parallel coordinates between age group, region, and games attended

- there's actually no trends, like what the hell is this dataset

```
In [190... parallel_df = fan_merch_merged[['Customer_Age_Group', 'Customer_Region', 'Ga

parallel_df['Age_Code'] = parallel_df['Customer_Age_Group'].astype('category')
parallel_df['Region_Code'] = parallel_df['Customer_Region'].astype('category')

fig = px.parallel_coordinates(
    parallel_df,
    dimensions=['Age_Code', 'Region_Code', 'Games_Attended', 'Purchased_Merch'],
    color='Games_Attended',
    labels={
        'Age_Code': 'Age Group',
        'Region_Code': 'Region',
        'Purchased_Merch': 'Purchased Merch',
        'Games_Attended': 'Games Attended'
    },
    color_continuous_scale='Viridis'
)

fig.show()
```

```
In [191... parallel_df.head()
```

```
Out[191...
   Customer_Age_Group  Customer_Region  Games_Attended  Purchased_Merch  Age_
0          26-40          Canada          4          False
1          26-40          Canada          7          False
2          18-25          Canada          4          False
3           60+          Canada          4          False
4           <18          India          4          False
```

```
In [ ]: revenue_by_member = fan_merch_merged.groupby('Member_ID')['Unit_Price'].sum()
revenue_by_member
```

Out[]:

	Member_ID	Revenue
0	1370	0.0
1	7913	0.0
2	9041	0.0
3	13517	0.0
4	18474	0.0
...	...	...
69995	99977604	0.0
69996	99986747	0.0
69997	99991696	0.0
69998	99995269	0.0
69999	99999245	0.0

70000 rows × 2 columns